Game theory

IB 431 – Behavioral Ecology
Fall 2025
September 9th 2025

Outline

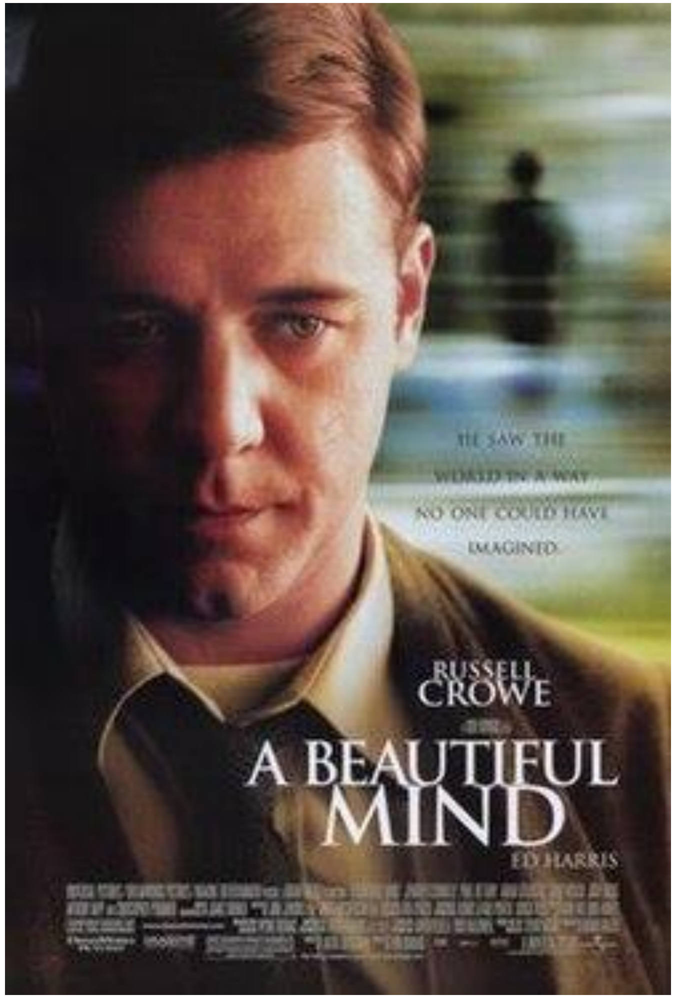
I. Overview of game theory
II. Card game
III. Fishing game
IV. Discuss Giraldeau et al 1994

Game theory

- Aims to build models to better understand decision making and strategic interactions
  - Analyzes how individuals or groups make choices when outcomes depend on the choices of others

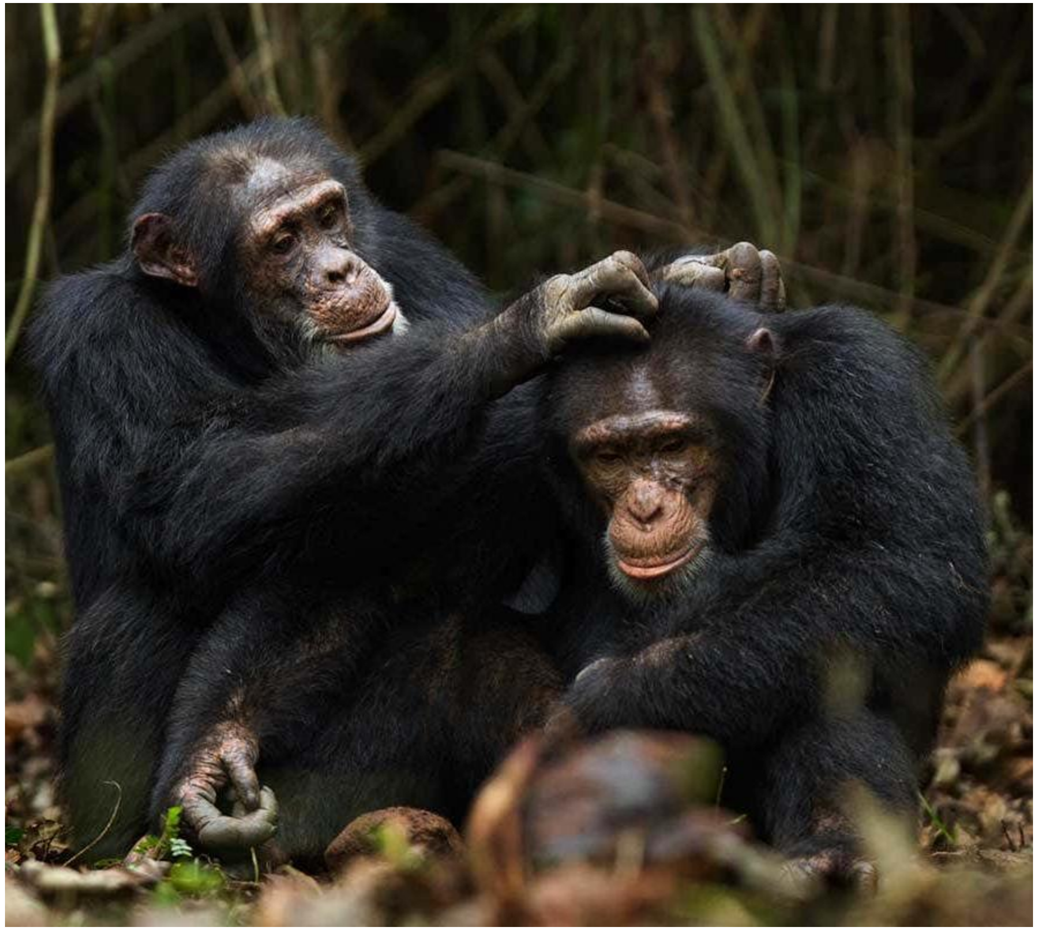
Game theory

- Aims to build models to better understand decision making and strategic interactions
  - Analyzes how individuals or groups make choices when outcomes depend on the choices of others
- Originated in the field of economics
  - John Nash (1994 Nobel Prize in Economics)
  - Nobel Prize in Economics has been about game theory 7 times



Applying game theory to animal behavior

- Framework for studying how animals make decisions
- Can better understand alternative strategies
- Particularly useful in study of:
  - Cooperation and competition
  - Social behavior and reproduction
  - Foraging
  - Communication



ELSEVIER

Animal Behaviour
Volume 36, Issue 2, April 1988, Pages 416-432

Callers and satellites in the natterjack toad: evolutionarily stable decision rules

Anthony Arak

Show more

Callers vs. satellites in natterjack toads

- Callers: croak to attract females
- Satellites: intercept females that respond to croaks
- Male toads adopted a frequency-dependent strategy which depends on their attractiveness to females

Game theory predicts an equilibrium

- Equilibrium occurs when payoff of one strategy equal to payoff of alternative strategy
- Evolutionary stable strategy (ESS)
  - Strategy or strategies where no outside strategy (e.g., from a mutation) will be able to propagate in the population
  - **For the toads: ESS is equilibrium of callers/satellites**

Doesn’t have to just be two strategies…

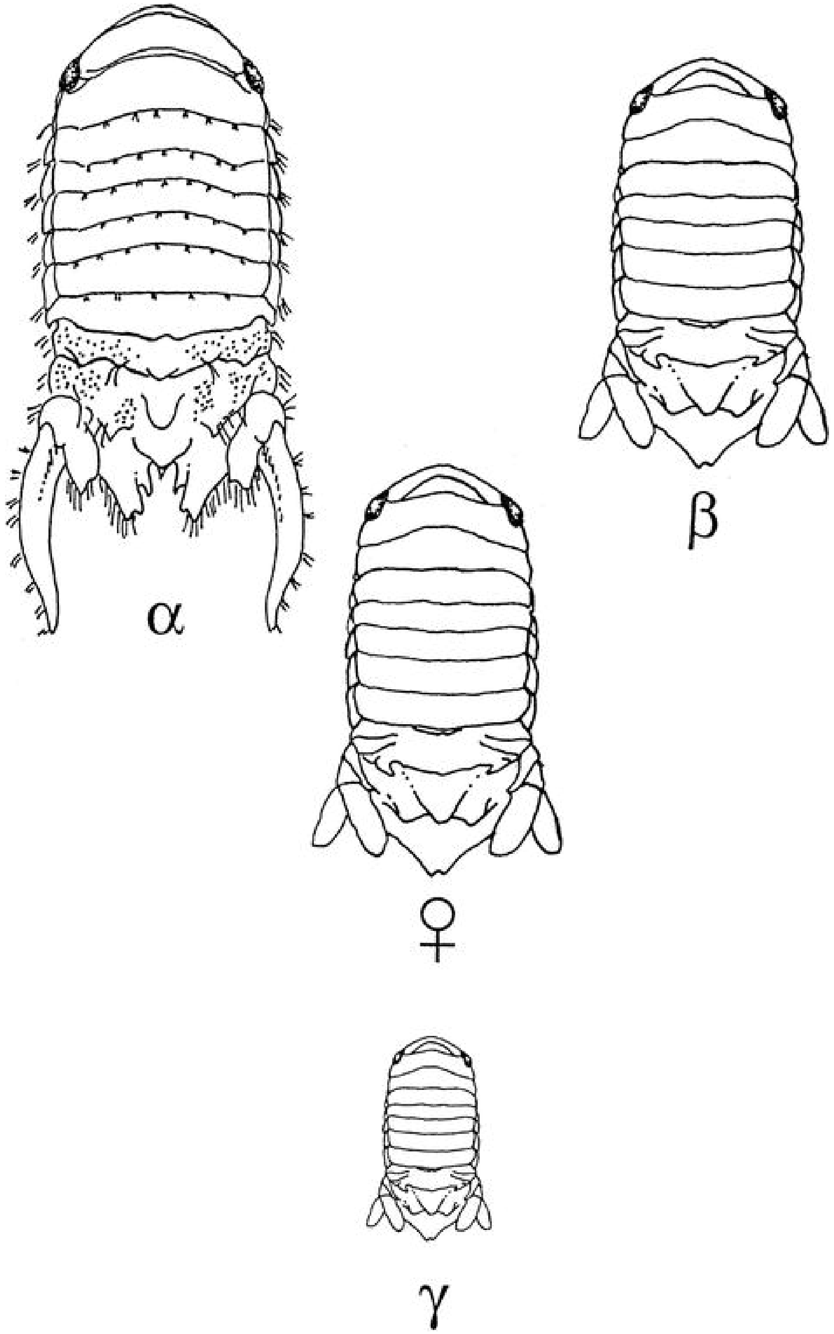
JOURNAL ARTICLE

Alternative Reproductive Behaviors: Three Discrete Male Morphs in *Paracerceis Sculpta*, an Intertidal Isopod from the Northern Gulf of California

Get access >

Stephen M. Shuster

Journal of Crustacean Biology, Volume 7, Issue 2, 1 April 1987, Pages 318–327,
https://doi.org/10.1163/193724087X00270
Published: 01 April 1987 **Article history** ▾



		Recipient’s fitness	
		Increases	Decreases
Actor’s fitness	Increases		
	Decreases		

		Recipient's fitness	
		Increases	Decreases
Actor's fitness	Increases	Cooperation	Selfishness
	Decreases	Altruism	Spite

Prisoner’s dilemma

- Everyone gets 1 red and 1 black card
- Find a classmate and play one of your cards
  - Both play red: both get 1 point
  - Both play black: both get 3 points
  - One plays red one plays black: whoever played red gets 6 points, whoever played black gets 0 points
- Repeat with 10 different partners

Card game

- <u>Both play red</u>: both get 1 point
- <u>Both play black:</u> both get 3 points
- <u>One plays red one plays black:</u> whoever played red gets 6 points, whoever played black gets 0 points
- Now play 10 times with the same partner

Different partners vs. same partner?

- Cheating generally best strategy
- Cooperation tends to be more likely if:
    - There are multiple interactions
    - Individuals can recognize and remember one another
    - Tit for tat strategy

Tit-for-tat in vampire bats

- Vampire bats will share blood meals with one another
  - Less likely to share with individuals that have not shared with them

Fishing game

- Each table is starting a fishing business
- Goal for each group is to **catch as many fish as possible**
  - Each 100 fish sell for 10¢
  - Fish population starts at 1000
  - Each group puts in a ‘claim’ for some percentage of fish
  - Surviving fish population doubles
  - Play for 8 time periods or until all fish claimed

	1	2	3	4	5	6	7	8	Profit
Fish pop.	1000								
Table 1									
Table 2									
Table 3									
Table 4									
Table 5									
Table 6									
Table 7									
Table 8									

Outline

I. Overview of game theory
II. Card game
III. Fishing game
IV. Discuss Giraldeau et al 1994